LMC 6650 Experiential Data Visualization
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Active Filtration

[https://colonel-snow.github.io/]

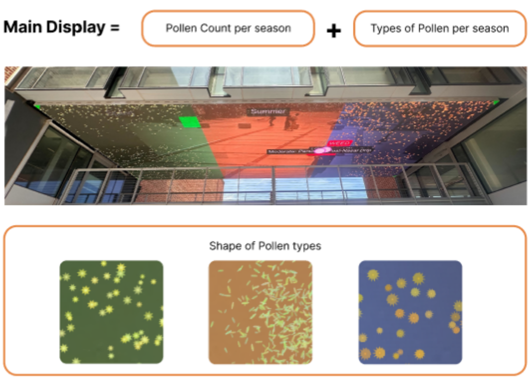
"How can pollen affect your life?"

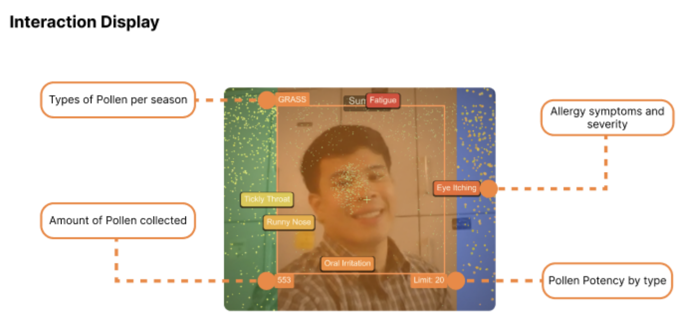
As the audience approaches the screen and their face appears in the camera, they should have pollen attached to their face as they walk past the screen. The display is divided into three vertical panels, each representing an allergy season: Spring on the left, Summer in the center, and Fall on the right. Pollen particles drift downward across all three panels simultaneously; each type rendered according to its real botanical characteristics, including weight, size, and shape derived from documented botanical data. As the audience stand in front of the screen, the pollen begins moving toward their face. Each panel simulates the behavior of the pollen type that peaks most prominently during that season

The particle density in each panel is not fixed. It is driven by real pollen count data pulled for that specific day in Atlanta, GA. On a high-count day, the threshold of pollen counts also increases. On a low-count day, the threshold of pollen counts also decreases. This means the experience changes depending on that specific day.

The audience can swipe a hand across the screen to disperse the pollen. It scatters on contact. Within seconds, it returns. This reflects a documented pattern among allergy sufferers: avoidance is partial and temporary. Pollen concentrations outdoors can reach tens of thousands of grains per cubic meter on high-count days, making complete exposure avoidance effectively impossible. When enough pollen accumulates around the face, the screen begins displaying symptoms specific to that allergen type.

The experience makes visible something that is normally invisible: the density, persistence, and biological specificity of the air breathed during allergy season.





Data Used

- Pollen Count per season
- Types of Pollen per season
- Pollen Potency by type
- Shape of Common Pollen Types
- Common allergy symptoms and severity

Data Sources:

- https://www.atlantaallergy.com/pollen_counts/
- https://www.atlantaallergy.com/pollen_counts/
- https://www.webmd.com/allergies/what-to-know-about-pollen-count
- https://www.sciencenews.org/article/physics-explains-how-pollen-gets-its-stunning-diversity-shapes
- https://www.cdc.gov/climate-health/php/effects/allergens-and-pollen.html#:~:text=Increased%20pollen%20effects,%2C%20runny%20nose%2C%20and%20congestion